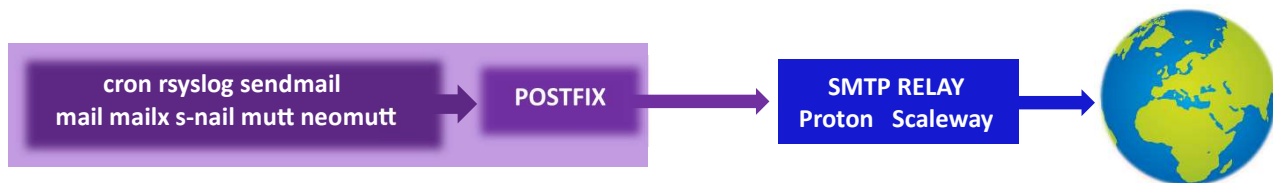


Guide for installing, configuring and using Linux POSTFIX to use Proton and Scaleway SMTP mail services

This guide describes how setup up Postfix to work with Proton Mail and Scaleway Transactional Email (TEM) secure SMTP services. Postfix when configured to work with a SMTP relay provides system and application administrators to use native and application mail tools to send reports, alerts and other information to external email addresses. The SMTP services of Proton and Scaleway are different in configuration requirements because they are designed for different workloads.



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Exclusions from this guide

- Specific software installation instructions are not included because software package management varies between Linux varieties.
- Configuration of internet domain record – refer to domain registrar documentation.
- Log management and rotation. Recommended if postfix is logging is enabled.
- The administration of local Linux mail, see **Local Linux mail management** section.
- The system logger, *journald*, cannot natively send mail alerts, journald relies upon scripts and third-party tools to send mail.

The realm of the illustrative participants

Mothra owns the domain name *infant.isle*, the Elias also use her domain and share the same Linux computer named *rolisica*.

Mothra aka *Queen of the Monsters* and *Mosura*, her home is Infant Island, she accompanied by the Elias (fairies) Moll, Lora and the distant Belvera. Mothra's destiny is to save the world from Godzilla, the inhabitant of Bass Island. Infant Island was subjected to nuclear testing by the nuclear power Rolisica.

Overview

All failed delivery notifications and mail misconfigurations result in mail accumulating in the local Linux user's mailbox; this has the potential to lead serious problems if local Linux mail is not managed. A filesystem filling up could result in catastrophic failure of applications and the device.

For this guide postfix 3.10.6 was used on Ubuntu 26.04 LTS Server. Both Proton and Scaleway can share the same domain name, see **Notes** section.

Basic Configuration

The configuration that is commonly found on the internet and transactional email vender supplied documentation.

Proton

- All mail has to come from a single sender, the sender address on the SMTP token.
- Cron only sends mails to the local Linux user mailbox as \$USER@localhost.
- Rsyslog will only send mail from the STMuser address

Scaleway

- Mail can be sent from individual email addresses.
- Display Names can be assigned as required.
- Cron mail is operational only if the MAILTO variable is set to a valid receiver. The Display Name cannot be set. If MAILTO not set, cron defaults to local Linux mail.
- Rsyslog ommail mail is fully operational if the mailfrom and mailto parameters are set to valid addresses. The Display Name cannot be set.

Default sender addresses (*Address rewriting – canonical maps*)

Senders do not have to provide a return address. These only apply to Linux users in the sender_canonical file.

Proton

- All mail has to come from the sender address on the SMTP token.
- Display name can be set as required.
- Cron and rsyslog mail fully operational, display name cannot be set, all mail from SMTPuser address.

Scaleway

- Provides full functionality and flexibility.
- Cron mail fully operational, will not default to local Linux mail.
- This feature is not required by rsyslog ommail because it adds no value.

Preconfigured sender addresses (*Sender-Dependent SASL authentication*) Only required for Proton STMP

In this configures multiple individual users to use a single Proton custom domain. This feature is not required for Scaleway. Selected users no longer required to come from the same SMTPuser.

- Each additional sender address has to be configured in Proton
- Each additional sender requires a STMP token for the same domain.
- Cron, use **sender_canonical**, if sender_canonical not enabled, cron can only send mail to the local Linux user.
- Rsyslog mail can use preconfigured email addresses.

This feature is also used to configure postfix to use multiple domains.

See **Test** section for more information on the basic and other configuration options. Results may vary in different Linux varieties and configurations.

Records of SMTP mail activity

Proton – Records accessed in the mail account that has the custom domain configured.

- Every mail processed is added in full to the **Sent** folder of the Proton account.
- Delivery failures are sent to the account In Box.

Scaleway TEM Essential – Records accessed in web account console, no download facility.

- For every mail processed, the delivery status, sender, receiver, subject and date-time are recorded.
- Mail records can be searched, filtered for failed deliveries and other events.
- Summary reports of mail activity are regularly mailed to the Scaleway account owner.

SMTP user credentials

Obtain SMTP access information from Proton and Scaleway, Username and Key will look like these:

	Proton Mail	Scaleway
Server	smtp.protonmail.ch	smtp.tem.scaleway.com
Port:	587	
Username	mothra@infant.isle	1d4b4e1d-0d11-406e-87d6-fcd11f7a8c68
Key/Token	1B9B4ZTG23Y50X2	8371c9b4-fc23-50c2-8769-74c7cb669769

Every Proton SMTP user address has to be removed from the Proton Contact List.

Postfix installation and configuration

As root, install the **mailutils** package. This include sendmail, mail, mailx and other mail utilities. Ubuntu bundles postfix with mailutils. If postfix is not bundled with mailutils install that after mailutils.

The postfix installation requires the user to enter their System mail name with is the custom domain name.

Set mail configuration type to
Internet Site

Postfix Configuration

Please select the mail server configuration type that best meets your needs.

No configuration:
Should be chosen to leave the current configuration unchanged.

Internet site:
Mail is sent and received directly using SMTP.

Internet with smarthost:
Mail is received directly using SMTP or by running a utility such as fetchmail. Outgoing mail is sent using a smarthost.

Satellite system:
All mail is sent to another machine, called a 'smarthost', for delivery.

Local only:
The only delivered mail is the mail for local users. There is no network.

General mail configuration type:

- No configuration
- Internet Site**
- Internet with smarthost
- Satellite system
- Local only

<Ok> <Cancel>

The 'mail name' is the domain name used to 'qualify' from <root>: please do not make your machine send

This name will also be used by other programs. It

Thus, if a mail address on the local host is foo@e

System mail name:

infant.isle

<Ok>

Set **System mail name** to the
your domain name

Next

As the root user continue and go to the **/etc/postfix** directory to configure postfix.

Make a backup copy of the original **/etc/postfix/main.cf** file before any changes.

Edit the main.cf

The system mail name(**infant.isle**) and hostname (**rolisica**) are added automatically during the installation process.

Remove all the comment lines, if required they can be referenced in the original main.cf backup.

Only add or change the highlighted lines.

```
compatibility_level = 3.9
myorigin = /etc/mailname
smtpd_banner = $myhostname ESMTP $mail_name (Ubuntu)
inet_protocols = all
mynetworks_style = host
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mydestination = rolisica, infant.isle, rolisica, localhost.localdomain, localhost
mailbox_size_limit = 0
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
biff = no
recipient_delimiter = +
relayhost = [smtp.protonmail.ch]:587
#relayhost = [smtp.tem.scaleway.com]:587
cyrus_sasl_config_path = /etc/postfix/sasl
smtpd_tls_key_file = /etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_cert_file = /etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_security_level = may
smtpd_tls_CAfile = /etc/ssl/certs/ca-certificates.crt
smtpd_tls_security_level = encrypt
smtpd_tls_session_cache_database = btree:${data_directory}/smtpd_scache
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated
defer_unauth_destination
myhostname = rolisica
inet_interfaces = loopback-only
sender_canonical_maps = hash:/etc/postfix/sender_canonical
smtp_sasl_auth_enable = yes
smtp_sasl_security_options = noanonymous
smtp_sasl_password_maps = hash:/etc/postfix/sasl/sasl_passwd
#maillog_file = /var/log/postfix.log
#debug_peer = smtp.protonmail.ch
```

The original mydestination generated at postfix installation, replace \$myhostname with actual hostname (rolisica)

```
mydestination = $myhostname, infant.isle, rolisica, localhost.localdomain, localhost
```

Cron and rsyslog mail will fail if **mydestination** is in this format:

```
mydestination = rolisica, localhost.infant.isle, localhost
```

The following have to added or edited in the /etc/postfix/mail.cf file.

Parameters	Initial Setting	Required Setting
relayhost	<i>Not set</i>	[smtp.protonmail.ch]:587 OR [smtp.tem.scaleway.com]:587
smtpd_tls_security_level	<i>May</i>	Encrypt
inet_interfaces	<i>All</i>	loopback-only
smtp_sasl_auth_enable	<i>New</i>	Yes
smtp_sasl_security_options	<i>New</i>	noanonymous
smtp_sasl_password_maps	<i>New</i>	hash:/etc/postfix/sasl/sasl_passwd
sender_canonical_maps	<i>New</i>	hash:/etc/postfix/sender_canonical
maillog_file*	<i>New</i>	/var/log/postfix.log
debug_peer*	<i>New</i>	smtp.protonmail.ch

Run **postfix reload** to make alteration to the main.cf effective.

* Only enable logging and debugging when required.

Debugging output is written to the postfix.log if enabled, otherwise the /var/log/mail.log file.

Only root to have read/write permissions to /var/log/postfix.log file.

Parameters	Purpose
mydestination	The destination for postfix mails.
relayhost	
smtp_tls_security_level	Encrypt outgoing mail
inet_interfaces	<i>loopback-only</i> blocks the receiving of external mail.
smtp_sasl_auth_enable	Configuration for the use of SASL authentication.
smtp_sasl_security_options	
smtp_sasl_password_maps	
sender_canonical_maps	This parameter permits root, applications and local accounts to send mail.
maillog_file	Trouble shooting and security forensics.
debug_peer	

Next...

Enabling SASL security

Continuing as the root user, go to **/etc/postfix/sasl** directory, it will be empty. Create a **sasl_passwd** file with the following contents

```
[SMTP Server]:<SMTP Port> <SMTP User name>:<SMTP Key>
```

Using credentials from mail providers the contents will be something similar to this.

```
[smtp.protonmail.ch]:587 mothra@infant.isle:1B9B4ZTG23Y50X2
```

or

```
[smtp.tem.scaleway.com ]:587 1d4b< SMTP Username>a8c68:8371<Secret Key>c7cb669769
```

Run postmap command to setup the SMTP user account and password for postfix

```
postmap /etc/postfix/sasl/sasl_passwd
```

This will create **sasl_passwd.db** file in the same directory as the **sasl_passwd** file.

By default, these files can be read by everybody, change them to root only readable.

Reload postfix configuration

```
postfix reload OR systemctl restart postfix
```

Basic setup is complete

Troubleshooting

To test the SMTP TLS connection the following should return something like this if everything is OK.

```
$ echo "q" | openssl s_client -connect smtp.protonmail.ch:587 -starttls smtp # OR
$ echo "q" | openssl s_client -connect smtp.tem.scaleway.com:587 -starttls smtp
Connecting to 51.159.84.239
CONNECTED(00000003)
depth=2 C=US, O=Internet Security Research Group, CN=ISRG Root X1
verify return:1
--- < plus a lot more >---
```

Postfix and system logs

- If no log management is in place, only enable postfix logging and debugging only when required. Any change to the main.cf file requires a **postfix reload**
- Logging will record the local Linux account that sends mail and the success and failure of each mail.
- Debugging output will go to **/var/log/mail.log** if logging is not enabled or **/var/log/postfix.log** if enabled.
- Debugging with default set produces substantial output for every mail sent.
- Additional information can be found in the systemd journal (log)

```
journalctl -f -t postfix/smtpd # smtp specific info
```

Setting default Linux user mail addresses

aka *Address rewriting – canonical maps*

To enable system mails from root and applications where the *From* address cannot be assigned at time of mail creation. The From address can be preconfigured using the **sender_canonical** facility.

Create a **/etc/postfix/sender_canonical** file. The format of the file is two columns, the first is the local user name and the second is the address that postfix will rewrite it to. A Scaleway mail Linux user that is in the sender_canonical file can override their default email address.

Proton

```
# <local_user> < user@domain>
root mothra@infant.isle
morusra mothra@infant.isle
queenie mothra@infant.isle
moll mothra@infant.isle
lora mothra@infant.isle
```

All mail has to come from the SMTP user
mothra@infant.isle

Scaleway

```
# <local_user> < user@domain>
root noreplay@infant.isle
morusra mothra@infant.isle
queenie mothra@infant.isle
moll elias@infant.isle
lora elais@infant.isle
```

All mail has to come from the domain
@infant.isle

No restriction on individual senders

Every time sender_canonical file is edited the map has to be compiled and reloaded into Postfix with the command

```
postmap /etc/postfix/sender_canonical
```

This will create **sender_canonical.db** in the **/etc/postfix** directory. Next

Reload/restart postfix

Scaleway configuration is complete

Enabling preconfigured sender email addresses – Proton SMTP

aka *Sender-Dependent SASL authentication*

The **sender_relay** feature has many uses, this configuration permits a user to use a preconfigured sending email address. The preconfigured email addresses can be shared between Linux users.

Sending of all cron mail fails because there is no method setting the sender as one of the sender_relay addresses or the primary domain address (mothra@infant.isle). This can be enabled using the **sender_canonical** facility. Rsyslog can send mail from any address in the sasl_passwd file.

Proton prerequisites

- Create a Proton Mail address for every Linux user that requires this facility.
- Create a Proton SMTPuser token for each address.
- If the user already exists, remove them from the Contact List

Add each address and token to the **/etc/postfix/sasl/sasl_passwd** file in this format

```
defcom1@infant.isle defcom1@infant.isle:38GHVEHZVEHC7RMB
defcom2@infant.isle defcom2@infant.isle:ZVEHC738GHVEHZVE
defcom3@infant.isle defcom3@infant.isle:VEHZVEZVEHC738GH
noreply@infant.isle noreply@infant.isle:DY4WHPY5ALY64FGY
morusra@infant.isle morusra@infant.isle:Y5ALY38GHVEHC7NB
queenie@infant.isle queenie@infant.isle:UEAMJK38GHVEHZC9
elias@infant.isle elias@infant.isle:8GHVEHC7NBUEAMJK
[smtp.protonmail.ch]:587 mothra@infant.isle:9UT8UXRMB9AYGRTY
```

Email addresses for defcom1, defcom2 and defcom3 are broadcast email addresses

In the `/etc/postfix` directory create a **sender_relay** file and addresses in this format

```
defcom1@infant.isle [smtp.protonmail.ch]:587
defcom2@infant.isle [smtp.protonmail.ch]:587
defcom3@infant.isle [smtp.protonmail.ch]:587
noreply@infant.isle [smtp.protonmail.ch]:587
morusra@infant.isle [smtp.protonmail.ch]:587
queenie@infant.isle [smtp.protonmail.ch]:587
elias@infant.isle [smtp.protonmail.ch]:587
```

Update postfix lookup tables. This will create a **sender_relay.db** file

```
postmap /etc/postfix/sasl/sasl_passwd /etc/postfix/sender_relay
```

Add the following to the `/etc/postfix/main.cf` file

```
smtp_sender_dependent_authentication = yes
sender_dependent_relayhost_maps = hash:/etc/postfix/sender_relay
```

Reload/Restart postfix

Usage

The default Display Name is the Linux user account name.

```
echo "Tokyo ETA 45 mins" | mail -s "Danger Godzilla" -r defcom2@infant.isle \
-- mothra@infant.isle
```

The Display Name can be altered as required.

```
echo "Tokyo ETA 45 mins" | mail -s "Danger Godzilla" -a "From: Moll defcom2@infant.isle" \
-- mothra@infant.isle
```

Configure cron to send email

The address format "Display Name <mothra@infant.isle>" is not permitted in cron and rsyslog

Cron requires the environment variable MAILTO to set.

If MAILTO not set will default to either local mail or the cron user default email address.

```
# Cron resets MAILTO environment variable when it scans the crontab file from top to bottom.
MAILTO=sysadmins@overthere.com
0 3 * * 0 /usr/local/bin/weekly-reports

# Reset MAILTO somebody else, the backup crew
MAILTO=bkups@overthere.com # Reset MAILTO to the backup crew
0 20 * * * /usr/local/bin/do-some-bkups

# Disable cron mail for the rest of crontab
MAILTO=""
* 1 * * * /usr/local/bin/doit-1 > /dev/null 2>&1 # Send stdout and stderr to the bit bucket

# This is not a test of cron sending an email
* 8 * * * /usr/local/bin/doit-2 | mail -s "Its done" -r mothra@infant.isle queenie@infant.isle
```

- Multiple recipient list format: `MAILTO="mothra@infant.isle,elias@infant.isle"`
- Some Linux varieties have a MAILFROM environment variable.

Configure rsyslog to send email

Rsyslog uses the ommail module of rsyslog.

As root create the file `/etc/rsyslog.d/alert.conf` with the following contents. Enter mail `body template` as a single line.

```
module(load="ommail")
template (name="mailBody" type="string" string="From %hostname%\nTimestamp :
%timereported%\nSource      : %programname%\nSeverity   : %rsyslogseverity-text%\nMessage
:%msg%")

template (name="mailSubject" type="string" string="%hostname% - Rsyslog alert for Program -
%programname%")

# Send on severity 0 Emergency, 1 Alert and 2 Critical
if $rsyslogseverity <= 2 then {
    action(type="ommail" server="127.0.0.1" port="25"
        mailfrom="rsyslog@localhost"
        mailto="root@localhost"
        subject.template="mailSubject"
        body.enable="on" template="mailBody"
        action.exeonlyonceeveryinterval="60") # Tests: set for 1 min
}
```

Restart rsyslog.

- Run **systemctl status rsyslog** to validate the alert.conf file, all content and format errors will be reported
- Replace mailfrom and mailto addresses for external recipients.
- Multiple recipient list format: mailto =["mothra@infant.isle","elias@infant.isle"]

Tests

Recommend initial testing to use the above configuration which send mail locally to root if postfix has been configured correctly.

From any user, run logger command to create system alerts. This will trigger rsyslog to mail alert messages if the logger severity is less than defined in alert.conf file. The smaller the number the more serious the alert is.

```
logger -t Test-tag -p user.alert "Testing ALERT level alert, 1 severity"
logger -t Test-tag -p user.crit "Testing CRITICAL level alert, 2 severity"
```

If logger sends an emerg (0) severity, systemd *will broadcast the emergency message to every user on the server.*

Notes

- Logger **-p** : severity levels:
 - 0 (emerg) Emergency
 - 1 (alert) Alert
 - 2 (crit) Critical
 - 3 (err) Errorthere are eight severity levels
- Validate logger alerts `journalctl -p err` will list all alerts with severity level <= 3. The user has to be a member of the journal, adm or wheel group depending upon the variety of Linux.
- The spam control parameter **action.exeonlyonceeveryinterval= <seconds>** defines how often to poll rsyslog, set to longer periods to prevent spam (20 mins = 1200 , 1hr 3600 etc)
- Alternative trigger examples

```
if $syslogseverity <= 3 and not ($msg contains "something not interesting") then {
    if $msg contains "something interesting " then {
```


Local Linux mail management.

Automated mailing and manual command line mailing requires monitoring and management of logs plus the mail queues and the mailboxes in the /var/mail directory.

Non-delivery reports and other error mails accumulate in user mailboxes, which could cause catastrophic failure of applications and server functionality if a filesystem filled up.

The *mail* utility can run interactively to remove mails in bulk (d *). It is safe to delete root and any other uses mail file in the /var/mail/ directory. A new file will be created when a mail arrives for that user.

Mails with invalid options that are discarded by postfix may not appear in the Linux user's mail box. If postfix logs are enabled, check them for "rejected" mails.

Tests

Basic setup

- Don't forget to check your spam/junk folder;
- There may be *some variation depending upon your Linux variety*.

For Proton and Scaleway

```
date | mail -s "Test from $USER" -r mothra@infant.isle -- godzilla@bass.isle
```

For Scaleway only - Setting display name using "Display Name <sender_address>" standard format.

```
date | mail -s "Test " -a "From:$(uname -n) <$USER@mothra.isle>" -- godzilla@bass.isle
```

- This test will fail for Proton. Proton will only accept mail the SMTPuser any display will either be ignored or fail. In this test the display name has to be one word. Proton expects all mail to come from the SMTP user with no Display Name, this test will fail with mail being rejected.
- The "--" separates the recipient addresses from other parameters, it is optional.
- Multiple recipients are a comma delimited list

Cron

If the cron MAILTO environment variable is not set all mail is delivered via postfix to the cron users local mailbox

If MAILTO is set to a valid external email address:

- *Proton* - It will fail because it does not come from the SMTPuser. Mail is sent from the cron user's account
- *Scaleway* – It will succeed

See section: **Setting default Linux user mail addresses**

Rsyslog

- *Proton* – rsyslog will send mail only mailfrom is set to the SMTPuser.
- *Scaleway* – fully functional, mailfrom can be rsyslog@domain.name

Default user email addresses.

Test 1 -The **mail -a** and **-r** options are no longer required

```
date | mail -s "Test from $USER" -- godzilla@bass.isle
```

- ◆ Success for both Proton and Scaleway
- ◆ The Display Name is the Linux user account name in Proton and Scaleway.

Test 2 Adding a **Display Name** to the default email address of the sender.

```
date | mail -s "Test from $USER" -a "From: Queenie mothra@infant.isle" godzilla@bass.isle
```

- ◆ Success for both Proton and Scaleway

Test 3 Over-riding the default email address of the sender adding a Display Name.

The default sender email address for Moll and Lora is elias@mothra.com. The Elais is name given to the fairies.

```
date | mail -s "Test from $USER" -a "From: Lora fairies@infant.isle" -- godzilla@bass.isle
```

- ◆ Scaleway succeeds, there are no restrictions on the individual sender i.e. *<anybody>@infant.isle*
- ◆ Proton only succeeds if Lora is sending it from her account and uses her Linux account name (lora), "From: Lora lora@infant.isle" is rewritten to the default as "From: Lora mothra@infant.isle".
See section **Enabling multiple senders from Linux** to fix this problem

Test 4 Cron

- ◆ Proton
If the environment variable MAILTO is set, all mail will succeed with the sender being the SMTPuser (mothra@infant.isle), no Display Name.
If MAILTO is not set cron mail defaults to local mail.
- ◆ Scaleway
If MAILTO is set, all mail will succeed with the sender being the noreply@domain_name (noreply@infant.isle), no Display Name. The noreply being inserted by system mail.
If MAILTO is not set, cron mail defaults to <hostname>@domain_name. If not a a valid address, it will fail.

Test 5 Rsyslog – Tests results are the same as **Basic setup** tests, previous page.

Notes

- Additional testing facilities
 - ◆ <https://dnschecker.org/all-tools.php> - A collection of domain and other tools.
 - ◆ swaks – A Linux SMTP testing tool
- Both Proton and Scaleway SMTP services can share the same domain. Only use Proton MX records, do not use Scaleway MX records. Only one set is required and Proton is pedantic about having MX records and their required priority.

Some domain registrars will automatically append in the domain to some records after entered, if a record entry fails, remove the domain name.
- Cc and Bcc Do not work as expected in postfix, the default settings of postfix treat all mail addresses the same. If a mail from sendmail or mailx is feed into postfix any addresses assigned to **Cc** and **Bcc** will be processed as **To** addresses. Using **sender_bcc_maps** to enable Bcc has a scope of usage too narrow to make it useful.
- Check Linux variety for mail utility and configuration options, they vary. Ubuntu mailx does not have **-c** (Cc) and **-b** (Bcc) options and **-A** is used for attachments others use **-a**. Suse Linux mailx has the **-t** option it is absent in Ubuntu. Red Hat has replaced mailx with s-nail.
- In some Linux varieties cron has a MAILFROM environment variable option, Ubuntu does not have that option.
- From (-r) mail option addressing in the formats will be sent "No Reply <mothra@infant.idle>". The "No Reply" will be discarded, the recipient only sees *mothera@infant.isle* with the display name being the Linux user account name. If using -r option, best to go with mothra@infant.isle format with no decorations.

ommail: Mail Output Module

<https://rsyslog.readthedocs.io/en/latest/configuration/modules/ommail.html>

rsyslog Properties

<https://docs.rsyslog.com/doc/configuration/properties.html>

Postfix: sender-dependent SASL authentication

<https://gist.github.com/zmwangx/223b4a5a55eb46193cfb>

Example - sending mail

These are the basics for using a heredoc to send mail. This style is for sendmail without postfix, postfix will still send this type of message.

```
#/usr/bin/bash
#sudo /usr/bin/bash -c 'echo "===== $(date)===== " \ 1
>> /var/log/postfix.log'

function sys_storage(){
# Filesystem storage usage
echo "-----"
df -h | awk '{printf "%7s %7s %7s %7s %s %s\n", $2, $3, $4, $5, $6, $7}'
echo "-----"
df --total -h | grep total | awk '{printf "%7s %7s %7s %7s %s\n", $2, $3, $4, $5, "TOTAL"}'
}

recpt_addr="godzilla@bass.isle"
cc_addr="gigan@xilien.co" 2
bcc_addr=""

/usr/sbin/sendmail -i -r mothra@infant.isle -t <<- msg_end
To: $rec_addr 3 4 5
Cc: $cc_addr
Bcc: $bcc_addr
Subject: "$(uname -n)local mail capacity report $(date)" 6

Number of mails in the queue is .. $(mailq)
Number of mails in $USER mailbox . $(grep -c "^From " /var/mail/$USER)
Size of $USER mailbox ..... $(du -h --apparent-size /var/mail/$USER | awk '{print $1}')
Total size all mailboxes ..... $(du -h --apparent-size /var/mail/ | awk '{print $1}')

Files system Storage
$(sys_storage)

Produced by $USER 6
msg_end
```

1. During testing, time stamp the postfix log, this makes the test obvious. Logging must be enabled
2. The addresses in Cc: and Bcc: will be processed as To: addresses. Postfix doesn't discriminate; they are all the same.
3. The option -i suppresses of the "dot on a line by itself" problem. A solitary dot on a line will terminate body input and initiate the sending of the mail, problematic if a command or application produces that as an output line.
4. This may not be required, see sections **Setting default Linux user mail addresses** and **Enabling preconfigured sender email addresses**
5. -t is a requirement for sendmail to accept To, Cc and Bcc from STDIN.
<<- starts the heredoc, the optional "-" removes unwanted indentation, msg_end are the heredoc tags of your choice.
6. It is helpful to include information about the source of the mail in the subject or body. The sender address is not reliable.